

Hardcoded Call Categorization Schema

Source: `backend/src/services/geminiService.js:13-232` — the `CATEGORIZATION_SCHEMA` constant.

These 17 top-level categories and **196 sub-categories** are embedded directly in the Gemini analysis prompt and populate the `category` / `sub_category` fields on `call_analysis` and `calls`.

Not to be confused with `call_category` / `call_sub_category` — that is the *dynamic* taxonomy stored in the `call_categories` Mongo collection and regenerated by `POST /api/analysis/generate-categories` and `POST /api/analysis/generalise-categories`.

Last revised 2026-08-14 — 14 sub-categories added and 2 duplicates resolved, based on the UPTET email corpus. See `email-taxonomy-fit.md` for the evidence behind each change.

Summary

#	Category	Subs
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11	Login & Account Access	8
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16	Result & Merit List	8
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Total		196

Entries marked  were added in the 2026-08-14 revision.

Full listing

1. Portal Access & Registration

1. Fresh Registration Query
2. Duplicate Registration Concern
3. OTR vs Exam Application Confusion
4. Resume Incomplete Application
5. Offline Application Query
6. Portal Not Loading / Technical Error
7. Registration Form Submission Error
8. Multiple OTR Accounts Issue
9. OTR ID Not Received After Registration
10. Registration Confirmation Not Received
11. Apply / OTR Link Not Visible on Portal 

2. Identity Verification

1. Aadhaar OTP Not Received
2. Aadhaar Number Not Accepted
3. Aadhaar Mismatch
4. Name / DOB Mismatch Across Documents
5. Identity Proof Selection Query
6. Live Photo / Face Match Failure
7. Manual Aadhaar Verification Request
8. Photo Clicked During Verification Issue
9. Aadhaar Linked Mobile Not Available
10. Name Prefix Mismatch (KM / Kumari) 
11. Post-Marriage Surname Mismatch in eKYC 
12. Wrong ID Type Used at Registration 

3. OTP, Password & CAPTCHA

1. Mobile OTP Not Received
2. Email OTP Not Received
3. OTP Expired Before Use
4. Wrong OTP Entered Multiple Times
5. Password Forgotten / Reset
6. CAPTCHA Not Loading / Unclear
7. Account Recovery Query
8. OTP Coming on Wrong Number

4. Category & Reservation

1. General Category Query
2. OBC Category & Creamy Layer
3. OBC Non-Creamy Layer Certificate Query
4. SC / ST Category

5. SC / ST Sub-Category Clarification
6. EWS Category
7. EWS Certificate Format / Validity Query
8. Divyang / PwD / PH Category
9. Disability Type & Percentage Query
10. Dependent Freedom Fighter Category
11. Ex-Army / Ex-Serviceman Category
12. Age Relaxation Query
13. UP Residency & Reservation Eligibility
14. Category Certificate Date Validity Query
15. Category Change After Form Submission

5. Address & Personal Details

1. Permanent Address Entry Issue
2. Correspondence Address Entry Issue
3. District / State Dropdown Issue
4. Village / Ward / Tehsil Not Found
5. Pincode Not Accepted
6. Personal Details Correction Request
7. Date of Birth Entry Issue
8. Gender / Nationality Entry Issue
9. Twin Information Query
10. Father / Husband Name Entry Issue
11. Mobile / Email Change in Profile
12. Marital Status Field Query NEW

6. Educational Qualifications

1. Education Details Entry in OTR
2. Wrong Education Row Added
3. Board / University Not in Dropdown
4. B.Ed / D.El.Ed Qualification Entry
5. Graduation Subject / Stream Entry
6. Training Qualification Entry
7. Marks / Percentage Entry Issue
8. CGPA to Percentage Conversion Query
9. Year of Passing Entry Issue
10. Appearing / Passed Status Query
11. Final Year Appearing Candidate Entry
12. Multiple Degree Entry Issue
13. Appearing Option Missing in Dropdown NEW
14. Board Roll Number Digit Length Rejected NEW
15. Out-of-State / NIOS Qualification Not Listed NEW
16. Graduation Field Mandatory but Not Applicable NEW

7. Uploads & Documents

1. Photograph Upload Issue
2. Signature Upload Issue
3. Photo Identity Proof Upload
4. Academic Certificate Upload
5. Category / Caste Certificate Upload
6. Domicile / Residency Certificate Upload
7. Handwritten Declaration Upload Issue
8. File Size / Format Requirement
9. File Too Large Error
10. Blurry / Unreadable Document Rejection
11. Document Preview Not Showing
12. Photo Background Color Requirement
13. Photo Dimensions Not Accepted
14. Upload Button Not Working
15. Wrong Person's Document Uploaded 
16. Handwritten Declaration Language Requirement 

8. OTR Completion & Preview

1. Preview & Edit Before OTR Completion
2. OTR Profile Locked After Submission
3. Complete OTR Profile Step Query
4. OTR Submission Confirmation Not Received
5. Preview Section Data Missing or Wrong
6. How to Edit Saved OTR Data
7. OTR Final Submit Button Issue
8. Print / Download OTR Form

9. Exam Application & Eligibility

1. Paper I Eligibility Query
2. Paper II Eligibility Query
3. Both Papers Application Query
4. Subject Group / Combination Selection
5. Practising Government Teacher Details Entry
6. Qualification Status (Passed / Appearing)
7. B.Ed Appearing Candidate Eligibility
8. D.El.Ed / BTC / JBT Eligibility Query
9. Age Limit Eligibility Query
10. Exam Centre Preference Entry
11. Application Form Section Not Saving
12. How to Apply for Exam After OTR
13. In-Service Teacher Without Graduation Eligibility 

10. Payment & Fee

1. Fee Amount Query
2. Category-wise Fee Query

3. Both Papers Fee Query
4. Payment Gateway / Method Query
5. Net Banking / UPI / Debit Card Issue
6. Challan Payment Query
7. Payment Pending / Processing Status
8. Money Debited but Application Incomplete
9. Duplicate Payment Risk
10. Duplicate Payment Refund Query
11. Payment Reconciliation Request
12. Application Status Showing PAID Confirmation
13. Fee Receipt / Challan Download Issue
14. Fee Waiver for Reserved Category Query

11. Login & Account Access

1. Login Method Query
2. OTR ID Forgotten / Recovery
3. OTP Login Not Working
4. Account Locked / Blocked
5. Too Many Failed Login Attempts
6. Registered Mobile Not Accessible
7. Login with New Device Issue
8. Session Timeout Issue

12. Amendment & Post-Submission

1. Amendment Window Opening Date Query
2. What Fields Can Be Corrected
3. Amendment Process Step-by-Step Query
4. Correction Window Already Closed
5. Photo / Signature Amendment
6. Name / DOB Correction After Submission
7. Category Correction After Submission
8. Subject / Paper Change After Submission
9. Address Correction After Submission
10. Re-payment Required After Amendment
11. Amendment Confirmation Not Received
12. Wrong Exam Level Selected (Primary vs Junior) NEW
13. Form Cancellation / Withdrawal Request NEW

13. Exam Information

1. Important Dates & Schedule Query
2. Exam Pattern & Structure Query
3. Number of Questions / Total Marks
4. Qualifying Marks / Cut-off Query
5. Negative Marking Query
6. Exam Language / Medium Query

7. Question Paper Language Options
8. Exam Duration Query
9. Normalisation / Multi-Shift Query
10. TET Validity Period Query
11. Syllabus Query
12. Previous Year Paper Query

14. Admit Card & Certificate

1. Admit Card Release Date Query
2. Admit Card Download Process
3. Admit Card Not Downloading / Available
4. Wrong Details on Admit Card
5. Exam Centre / Date / Time Query
6. Exam Centre Change Request
7. TET Pass Certificate Download
8. TET Certificate Validity Query
9. DigiLocker Certificate Query
10. Photo Mismatch on Admit Card
11. Category Error on Certificate
12. Duplicate Certificate / Marksheet Query

15. Scribe & Compensatory Time

1. Scribe Eligibility Criteria Query
2. How to Request a Scribe
3. Scribe Arrangement Process
4. Scribe Documents & Declaration Required
5. Scribe Qualification / Education Limit
6. Compensatory Time (30 Minutes) Query
7. Scribe Declaration Form Submission
8. Medical Certificate for PwD Requirement
9. Disability Certificate Format Query

16. Result & Merit List

1. Result Declaration Date Query
2. Result Check Process
3. Merit List Query
4. Cut-off Marks Query
5. Rank / Score Discrepancy
6. District Allocation Query
7. Selection Process After TET
8. Waiting List Query

17. General Enquiry

1. Application Mode Query

- 2. Notification / Advertisement Query
- 3. Helpline Timing Query
- 4. Appointment / Job Guarantee Query
- 5. Transfer to Another Department
- 6. Call Back Request
- 7. Unrelated / Wrong Call
- 8. Repeated Call / Follow-up
- 9. Complaint Against Portal / Process

Change log

2026-08-14 — evidence-driven revision (184 → 196)

Derived from 1,895 UPTET helpline emails; every addition clears the ≥3-item evidence bar the schema's own generator prompt imposes ([geminiService.js:528-531](#)). Full analysis: [email-taxonomy-fit.md](#).

Duplicates resolved (–2). The schema previously violated its own mutual-exclusivity rule:

Sub-category	Was	Now
Password Forgotten / Reset	in both #3 and #11	#3 only — the category name explicitly claims "Password"
OTR ID Forgotten (#3) / OTR ID Forgotten / Recovery (#11)	same bucket, two names	#11 only , as OTR ID Forgotten / Recovery — identifier recovery is account access

The governing split is now: **#3 owns credential challenges** (OTP delivery, CAPTCHA, password), **#11 owns account access** (login methods, lockouts, identifier recovery, sessions).

[Account Recovery Query](#) remains in #3 and sits close to #11's [OTR ID Forgotten / Recovery](#). Not a duplicate — generic recovery vs. specific identifier lookup — but worth watching if the two start colliding in practice.

Sub-categories added (+14). Counts are matched rows in the 1,895-email corpus:

Rows	Added	To
33	Name Prefix Mismatch (KM / Kumari)	#2
26	Appearing Option Missing in Dropdown	#6
21	Post-Marriage Surname Mismatch in eKYC	#2
20	Apply / OTR Link Not Visible on Portal	#1
19	Wrong Exam Level Selected (Primary vs Junior)	#12
13	Wrong Person's Document Uploaded	#7
12	Out-of-State / NIOS Qualification Not Listed	#6
9	Board Roll Number Digit Length Rejected	#6
7	Handwritten Declaration Language Requirement	#7

Rows	Added	To
4	Graduation Field Mandatory but Not Applicable	#6
4	In-Service Teacher Without Graduation Eligibility	#9
3	Wrong ID Type Used at Registration	#2
3	Form Cancellation / Withdrawal Request	#12
3	Marital Status Field Query	#5

Together these capture **177 rows (9.3% of the corpus)** and lift schema coverage from **79.5% → 82.4%**.

The most consequential is **Appearing Option Missing in Dropdown**: it separates a *portal defect* from the eligibility-query bucket **Final Year Appearing Candidate Entry**, which alone was carrying 207 rows (11% of all traffic). The defect is now countable instead of hidden inside a query category.

Known issues

Returned values are not validated against the schema

`categorizeRecording` snaps hallucinated `call_category` / `call_sub_category` values back to safe defaults (`geminiService.js:1117-1140`), but `category` / `sub_category` receive only a cosmetic `cleanCategory()` prefix strip. There is no check that the returned pair actually exists in `CATEGORIZATION_SCHEMA`, so whatever Gemini emits is persisted as-is.

Two values bypass the schema entirely

Value	Written by	Condition
Call too Short	<code>analysisWorker.js:104-113</code>	Call duration > 0 and < 10 seconds — skipped before Gemini is called
Audio Unclear	Gemini, via prompt special case (<code>geminiService.js:906-920</code>)	Audio too noisy, silent, or unintelligible to identify the issue

Both are excluded from the summary corpus fed to `/generate-categories` (`analysis.js:549-553`).

These are **dispositions, not categories** — a distinction the schema does not yet model. See the disposition-axis recommendation in `email-taxonomy-fit.md`.

The schema is call-shaped

Call Back Request, **Unrelated / Wrong Call**, **Repeated Call / Follow-up**, and **Helpline Timing Query** are all phrased for voice calls and cannot match an email — which is why #17 General Enquiry scored zero hits across the entire email corpus. Unresolved; see the channel-neutrality recommendation in the analysis doc.